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ThreatConnect Playbook — Get Known Scanning Indicators by ASN (Non-CDN) Tagger

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1. Purpose

This ThreatConnect playbook automatically identifies and labels **non-CDN known scanning indicators** within HTOC's own indicator set. It queries for Address indicators belonging to the Google Cloud ASN (396982) that have not yet been labeled, applies the `Known Scanner PB` (**Known Scanner Playbook**) tag to classify them as known non-CDN scanning activity, and stores the indicator data in the ThreatConnect DataStore. Email alerts are sent on any failures.

The `Known Scanner PB` tag is the primary identifier that an indicator has been reviewed and classified by this playbook as originating from a known non-CDN scanning source. The TQL exclusion filter (`NOT hasTag("Known Scanner PB")`) ensures each indicator is only processed once. The playbook runs twice daily to keep labeling current as new indicators are added to ThreatConnect.

This playbook is the **non-CDN counterpart** to the CDN Tagger playbook (`Get CDN Scanning indicators by ASN Tagger.pbxz`), which handles ASN 63949 (Linode/Akamai). Together they cover both CDN and non-CDN scanning infrastructure.

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2. Scope

This procedure applies to HTOC analysts and ThreatConnect administrators who monitor, trigger, modify, or troubleshoot this playbook. It covers the playbook's trigger configuration, step-by-step logic, error handling, and maintenance guidance.

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3. Prerequisites

Requirement	Notes
ThreatConnect access	Must have playbook view/execute permissions
HTOC Org indicators	Address indicators with ASN 396982 must be present under the `HTOC Org` owner in ThreatConnect
ThreatConnect DataStore	Local DataStore must be available for POST operations
Email server configured	SMTPS (Implicit TLS) connection required for failure alert emails
Notification recipient	`jaytlin.askew@hhs.gov` — update if ownership changes

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4. Trigger Configuration

The playbook has three defined triggers. The **12 Hour Trigger** is the current active production trigger. The WebHook Trigger is used for testing and manual execution only.

Trigger Name	Type	Schedule	Status
12 Hour Trigger (6AM - 6PM)	Timer	6:00 AM and 6:00 PM (`0 0 6,18 * * ?`)	**Active — Production**
WebHook Trigger	HttpLink (Webhook)	On demand	Testing / manual use only

Trigger Name	Type	Schedule	Status
Every 2 Minutes Trigger (Test Trigger)	Timer	Every 2 minutes (`0 */2 * * * ?`)	Inactive

12 Hour Trigger (Production)

Type: Timer

Schedule: Cron `0 0 6,18 * * ?` — fires at **6:00 AM and 6:00 PM** daily

Purpose: Ensures new non-CDN scanning indicators are tagged and stored twice daily without manual intervention

WebHook Trigger (Testing / Manual)

Type: HTTP Link (external webhook)

Timeout: 5 minutes

Cache: 120 minutes (prevents duplicate executions within the cache window)

HTTP Response Body: Returns the success log message (`logger.content`) from the final success step

Purpose: Used for testing changes to the playbook or triggering a manual run outside the normal schedule. Send an HTTP request to the webhook URL found in the ThreatConnect playbook editor to invoke it.

> **Note:** Do not leave the WebHook Trigger as the primary active trigger in production. The 12-hour timer is the correct production trigger.

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5. TQL Query

The playbook queries ThreatConnect using the following TQL expression in the **ThreatConnect Get Indicators** step:

```
(ownerName in ("HTOC Org"))
AND typeName in ("Address")
AND addressASN IN (396982)
AND NOT hasTag(summary IN ("Known Scanner PB"))
```

Filter	Value	Purpose
Owner	`HTOC Org`	Limits to HTOC's own indicator set
Type	`Address`	IP addresses only
ASN	396982	Targets Google Cloud infrastructure (non-CDN known scanner range)
Tag exclusion	`NOT hasTag("Known Scanner PB")`	Skips indicators already processed by this playbook
Max results	5	Processes up to 5 indicators per execution
Fail on no results	No	Gracefully exits with no action if no new indicators are found

> **Important:** The `NOT hasTag("Known Scanner PB")` filter is what prevents reprocessing. If this tag is removed from an indicator, the playbook will process it again on the next scheduled run.

Monitored ASN

ASN	Known Provider
396982	Google Cloud

To add additional ASNs, update the `addressASN IN (...)` clause in the TQL field of the **ThreatConnect Get Indicators** step with a comma-separated list.

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6. Playbook Flow

6.1 Error Paths

Failure Point	Behavior
Get Indicators fails	→ Logger: "Get Indicators FAILED. Please check the TQL and summary for output result errors." → Email: subject "GET Indicator Process Failed", body = logger content, to `jaytlin.askew@hhs.gov`
No results found	Playbook exits cleanly — no email, no error (fail_on_no_results = false)
Iterator fails	→ Email: subject "Get CDN Scanning Indicators by ASN Failed.", to `jaytlin.askew@hhs.gov`
Set HTOC Scanner Identifier (tag add) fails	→ Logger: "Process Failed with an error." (ERROR) → Email: subject "Set HTOC Scanner Identifier Failed", body = TC error message, to `jaytlin.askew@hhs.gov` → Loop continues (does not halt remaining indicators)

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7. Step Details

Step 1 — ThreatConnect Get Indicators

App: TCPB - Threatconnect Get Indicators v1 (v1.0.9)

Action: TQL

Output: `tc.indicators.entity` (TCEntityArray) and `tc.indicators.summary` (StringArray)

Fail on no results: No — playbook exits cleanly if no untagged indicators are found

Also connects to: List of Indicators Summary Logger (logs the full summary list for each run)

Step 1a — List of Indicators Summary Logger

App: Logger (v1.0.0)

Level: INFO

Message: Contents of `tc.indicators.summary` (StringArray) — logs all indicator summaries returned by the query

Purpose: Provides a per-run audit log of which indicators were discovered before the loop begins

Step 2 — Iterator

App: Iterator (v1.0.0)

Input: `tc.indicators.entity` array from Step 1

Output: `currentIndicator` (TCEntity) — one indicator per loop iteration

Loop behavior: Iterates until all entities are processed; on loop completion (Pass) advances to the Success step; on failure sends an alert email

Step 3 — ThreatConnect Tags (Get)

App: TCPB - Threatconnect Tags TI v2 (v2.0.7)

Action: Get

Entity: `currentIndicator` from the Iterator

Fail on no results: No — indicators without existing tags proceed normally

Step 4 — ThreatConnect Set HTOC Scanner Identifier

App: TCPB - Threatconnect Tags TI v2 (v2.0.7)

Action: Add

Tag applied: `Known Scanner PB`

Apply to all: Yes

Entity: `currentIndicator`

Fail on error: Yes — failed tag additions route to the tag-failure email path

Purpose: Marks the indicator as a known non-CDN scanner and prevents reprocessing on future runs

Step 5 — JMESPath

App: TCPB - JMESPath v3.0 (v3.0.12)

Input: `tc.response.json.raw` from Step 4 (tag response JSON)

Expressions:

`[0].summary` → output variable `indicator` (String)

`[0]` → output variable `indicatorData` (String)

Fail on expression error: Yes

Step 6 — DataStore (POST)

App: TCPB - DataStore v4 (v4.0.5)

Action: POST

Domain: Local

Data type: JSON

Key (rid): `indicator` (the indicator summary from JMESPath)

Data: `{"indicatorData": <indicatorData from JMESPath>}`

Purpose: Persists indicator metadata to the ThreatConnect Local DataStore for downstream reference

Step 7 — Complete Success Logger

App: Logger (v1.0.0)

Level: INFO

Message: "Process completed with no issues"

Step 8 — Logger (Loop End)

App: Logger (v1.0.0)

Level: INFO

Message: "Tagging process completed."

Behavior: Triggers the EndLoop connection back to the Iterator

Step 9 — Success Logger

App: Logger (v1.0.0)

Level: INFO

Message: "Get CDN Scanning Indicators by ASN Completed Successfully!"

Output: `logger.content` — returned as the WebHook HTTP response body when triggered via webhook

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8. Email Alerts

All failure notifications are sent to `**jaytlin.askew@hhs.gov**`. Update this address in all three Send Email steps if the playbook owner changes.

Alert	Subject	Body	Trigger
Get Indicators failed	`GET Indicator Process Failed`	Logger error content	Get Indicators step fails
Iterator / no results	`Get CDN Scanning Indicators by ASN Failed.`	Static message	Iterator step fails
Tag add failed	`Set HTOC Scanner Identifier Failed`	ThreatConnect error message from tag step	Tag Add step fails

Connection type: SMTPS (Implicit TLS)

Timeout: 30 seconds

Send one: Yes (single email per event)

Fail on error: No (email failures do not halt the playbook)

> **Known issue:** The Iterator failure email subject reads "Get CDN Scanning Indicators by ASN Failed." — this is a copy-paste artefact from the CDN Tagger and does not affect functionality. Update the subject in the **Send Email** step connected to the Iterator's Fail path if clarity is needed.

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9. Modifying the Playbook

Adding or Removing Monitored ASNs

Open the playbook in the ThreatConnect editor.

Click on the **ThreatConnect Get Indicators** step.

Update the `addressASN IN (...)` clause in the TQL field.

Save and confirm the 12-hour trigger remains active.

Changing the Max Results per Run

`max_results` is currently **5**. Increase this to process more indicators per execution. With the 12-hour schedule this allows up to 10 indicators tagged per day at the default setting.

Changing the Scanner Tag Name

The tag `Known Scanner PB` is the marker used to prevent reprocessing. To rename it:

Update the `name` parameter in **ThreatConnect Set HTOC Scanner Identifier**.

Update the `NOT hasTag(summary IN ("Known Scanner PB"))` clause in the TQL.

Bulk-update existing tagged indicators in ThreatConnect before reactivating.

> **Warning:** If the tag name in the TQL exclusion filter is not updated to match the new tag being applied, indicators will be reprocessed on every execution.

Updating the Notification Email

Update the `recipient` field in all three **Send Email** steps to the new owner's address.

Managing Triggers

12 Hour Trigger (6AM - 6PM): This is the production trigger. It should always be active during normal operations.

WebHook Trigger: For testing and manual one-off runs only. Confirm the 12-hour timer remains the primary trigger.

Every 2 Minutes (Test Trigger): For development/debugging only. Disable immediately after use.

To toggle a trigger, open the playbook editor, click the trigger, and toggle the Active switch.

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10. Troubleshooting

Symptom	Likely Cause	Resolution
Email: "GET Indicator Process Failed"	TQL syntax error or API connectivity issue	Review the TQL in Get Indicators; verify ThreatConnect connectivity
No output, no email, no action	No untagged indicators found for ASN 396982	Expected behavior — `fail_on_no_results` is false; check ThreatConnect for new Address indicators under ASN 396982
Email: "Set HTOC Scanner Identifier Failed"	Tag add API error — possible permissions issue or indicator locked	Check ThreatConnect permissions for the playbook service account; review the TC error message in the email body

Symptom	Likely Cause	Resolution
Indicators being reprocessed repeatedly	`Known Scanner PB` tag is not persisting	Verify the tag add step is not failing silently; confirm the tag name in the TQL matches exactly (case-sensitive)
Iterator failure email says "CDN" in subject	Known copy-paste artefact from the CDN Tagger	Cosmetic only; update the Send Email step subject if needed
Playbook not triggering	12-hour timer inactive or misconfigured	Verify the 12 Hour Trigger is active in the playbook editor
DataStore POST errors	Local DataStore unavailable or data format issue	Check ThreatConnect DataStore status; verify JMESPath is producing valid `indicatorData` JSON

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11. Relationship to Other Playbooks

This playbook is one of a pair covering different scanning infrastructure categories:

Playbook	Tag Applied	ASN	Provider
Get Known Scanning indicators by ASN (none CDN) Tagger *(this playbook)*	`Known Scanner PB`	396982	Google Cloud
Get CDN Scanning indicators by ASN Tagger	`CDN PB`	63949	Linode / Akamai

Both playbooks share the same structure, run on the same 12-hour schedule, and write to the same Local DataStore. They differ only in the ASN they monitor and the tag they apply.

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12. Maintenance Notes

Tag name (`Known Scanner PB`): This is the production tag name used to classify non-CDN known scanners. Do not rename without simultaneously updating the TQL exclusion filter.

Max results = 5: Processes up to 5 new indicators per execution (10 per day with the 12-hour schedule). Increase if faster catch-up is needed.

fail_on_no_results = false: This playbook is designed to run silently when there are no new indicators to process, unlike the CDN Tagger which alerts on no results. This is intentional.

Active trigger is 12 Hour Timer: The production schedule runs at 6:00 AM and 6:00 PM. The WebHook trigger is for testing only.

Version history: This is version 1.13. Export a new `.pbxz` backup from ThreatConnect after any significant changes.

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13. Related Documents

`SOP_TC_Playbook_GetCDNScanningIndicatorsByASN.md` — CDN counterpart playbook SOP

ThreatConnect Playbook documentation: [ThreatConnect Playbook Apps](https://docs.threatconnect.com)